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Networked Influence in a Tabula Rasa Legislature

Co-authorship, Ideological Dynamics, and Political Success
in the Chilean Constitutional Convention (2021–2022)

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All the data, cleaned and fully linked — **soon public**



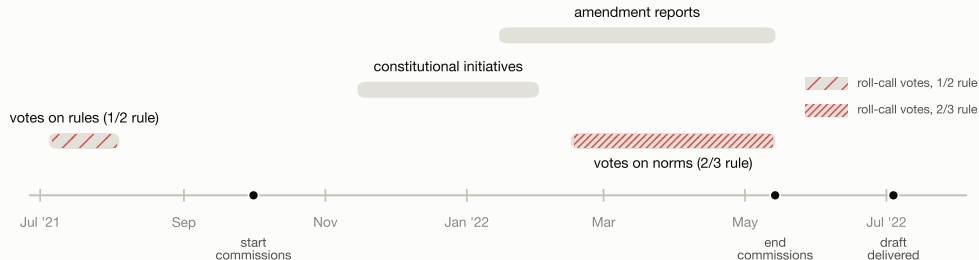
The data-cleaning repository

A Data in Brief paper announcing the release is in preparation.

A natural quasi-experiment

- The network you observe today is the sediment of decades of favors, committees, careers, etc.
- The Chilean Constitutional Convention (Jul 2021 – Jul 2022) started from **zero relational stock**: a majority of newcomers and independents, 17 reserved indigenous seats, a body that dissolved on delivery.
- They set their own rules:
 - Each constitutional initiative required **8–16 sponsors**.
 - Each norm needed **2/3 of the floor (103 of 154)** to be approved.

Chile's Constitutional Convention



The pipeline



What we know about each member

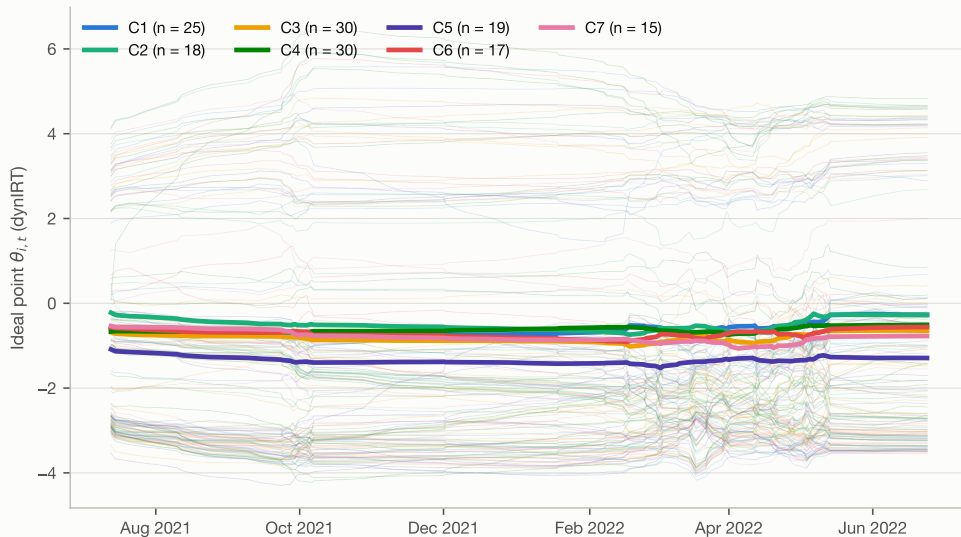
- **Political**: electoral list; district of election.
- **Profile**: age; gender; law degree; education level (HS / univ / master / PhD).
- **Prior institutional experience**: had held elective or appointed public office before the Convention (former MPs, mayors, officials) — 35 of 154 members.
- **Ideology**: two-dimensional ideal points from the first month of roll calls, *before* the collaboration network existed.

The seven thematic commissions

	Name (short)	Members	N lawyers	N exper.	Age	Education	Initiatives	Amendments	In draft
C1	Political system	25	15	9	42.2	1.44	89	365	51
C2	Const. principles	18	4	4	45.8	1.28	114	53	24
C3	Form of the state	30	11	12	45.1	1.43	51	241	67
C4	Fundamental rights	30	9	4	45.9	1.03	283	395	54
C5	Environment	19	4	3	44.7	1.26	133	559	30
C6	Justice systems	17	15	3	42.6	1.65	93	381	98
C7	Knowledge systems	15	2	0	51.7	1.33	64	332	35

- C4 (Fundamental rights): basically bachelor's degree, and 283 initiatives;
- C6 (Justice systems): 15/17 members were lawyers;
- C7 (Knowledge systems) has no politically experienced member.

Ideal-point trajectories by commission membership (thin lines = delegates, thick lines = commission means)



Research questions

Formation

- **RQ1**: can the co-sponsorship network be predicted from what people *brought with them* (district, lawyer, profile)?

Behavior

- **RQ2a**: does network exposure *move* members' ideological positions?
- **RQ2b**: does voting *defection* travel along co-sponsorship ties?

Success

- **RQ3a**: what makes an *article* survive into the draft?
- **RQ3b**: does the *context* an initiative is born into matter for its authors' success?

RQ1 — Formation

Could we predict the network?

Bipartite ERGM

- Unit of analysis: the **person - initiative** signature ($y_{ia} = 1$ if member i signed initiative a).

$$\Pr(Y = y) = \frac{\exp(\theta^\top s(y))}{\kappa(\theta)}$$

$$s(y) = \left(\underbrace{\sum_{i,a} y_{ia}}_{\# \text{ edges}}, \underbrace{\sum_a \left(\max_{i: y_{ia}=1} v_i - \min_{i: y_{ia}=1} v_i \right)}_{\text{range of } v \text{ per doc}}, \text{gwb1deg}, \text{gwdsp}, \dots \right)$$

- Variables: per-document ranges of θ_1, θ_2 , age, education; the bloc contingent's own θ_1 range; gwb1degree; gwdsp.
- Seven models, one per commission.
- Estimation: MPLE as a logistic on change statistics; SEs by **initiative bootstrap** ($B = 500$).

Same status, by bloc of the pair	C1	C2	C3	C4	C5	C6	C7
— pre-net: Lawyer							
Left	-0.07	0.06	0.02	0.06 ⁺	0.00	0.11*	-0.07
Center-left	0.11	0.11	0.13	-0.03	0.15*	-0.26	-0.00
Right	-0.11	-0.15**	-0.01	0.01	0.10	0.13*	-0.06
Reserved seats	0.06	0.24	0.10	-0.26 ⁺	0.20*	-0.21	-0.04
Others	0.12	-0.11	-0.09	0.09 ⁺	-0.00	-0.13	-0.15
Across blocs	0.06	-0.08*	0.02	0.04*	-0.05 ⁺	0.07 ⁺	0.06 ⁺
— pre-net: Prior institutional experience							
Left	0.12 ⁺	0.13*	0.09	0.11**	0.04 ⁺	0.06	0.00
Center-left	0.11	-0.03	0.09	0.09**	-0.09	-0.01	0.12
Right	-0.13	-0.14	-0.03	0.05	-0.14	0.00	-0.11
Reserved seats	0.06	0.00	-0.08	-0.05	-0.11	-0.10	-0.05
Others	-0.25	-0.26	—	0.05	—	-0.20	-0.30
Across blocs	0.04	0.06 ⁺	0.03	0.06**	0.04*	0.04	0.04
— pre-net: Gender							
Left	-0.12	-0.03	-0.06	-0.03	-0.02	-0.04	-0.09
Center-left	-0.08	-0.10	-0.07	0.00	0.02	-0.02	0.05
Right	0.14*	0.03	0.05	-0.06	0.04	-0.03	0.09
Reserved seats	-0.11	-0.50	-1.77	-0.60**	-1.64***	0.04	-0.26
Others	-0.21	-0.26	-0.61	-0.20**	-0.38*	0.19	-0.37
Across blocs	0.04	0.12**	0.20*	0.13***	0.12***	0.08 ⁺	0.11*
— pre-net: District							
Left	-0.64*	-0.48*	0.45	-0.45***	-0.29	-0.45*	-0.04
Center-left	-0.09	-0.80	-0.48	0.73*	0.18	-0.74	0.30
Right	1.00***	1.36***	0.42	0.41***	0.08	0.23	0.88*
Reserved seats	-0.07	0.13	-0.03	-0.27 ⁺	0.30	-0.03	0.19
Others	1.98 ⁺	—	-1.92**	—	-0.29	0.26	—
Across blocs	0.54***	0.28*	0.67**	0.39***	0.47***	0.34*	0.46**

Same status, by bloc of the pair	C1	C2	C3	C4	C5	C6	C7
pre-net: Lawyer							
Left	-0.07	0.06	0.02	0.06 ⁺	0.00	0.11*	-0.07
Center-left	0.11	0.11	0.13	-0.03	0.15*	-0.26	-0.00
Right	-0.11	-0.15**	-0.01	0.01	0.10	0.13*	-0.06
Reserved seats	0.06	0.24	0.10	-0.26 ⁺	0.20*	-0.21	-0.04
Others	0.12	-0.11	-0.09	0.09 ⁺	-0.00	-0.13	-0.15
Across blocs	0.06	-0.08*	0.02	0.04*	-0.05 ⁺	0.07 ⁺	0.06 ⁺
pre-net: Prior institutional experience							
Left	0.12 ⁺	0.13*	0.09	0.11**	0.04 ⁺	0.06	0.00
Center-left	0.11	-0.03	0.09	0.09**	-0.09	-0.01	0.12
Right	-0.13	-0.14	-0.03	0.05	-0.14	0.00	-0.11
Reserved seats	0.06	0.00	-0.08	-0.05	-0.11	-0.10	-0.05
Others	-0.25	-0.26	--	0.05	--	-0.20	-0.30
Across blocs	0.04	0.06 ⁺	0.03	0.06**	0.04*	0.04	0.04
pre-net: Gender							
Left	-0.12	-0.03	-0.06	-0.03	-0.02	-0.04	-0.09
Center-left	-0.08	-0.10	-0.07	0.00	0.02	-0.02	0.05
Right	0.14*	0.03	0.05	-0.06	0.04	-0.03	0.09
Reserved seats	-0.11	-0.50	-1.77	-0.60**	-1.64***	0.04	-0.26
Others	-0.21	-0.26	-0.61	-0.20**	-0.38*	0.19	-0.37
Across blocs	0.04	0.12**	0.20*	0.13***	0.12***	0.08 ⁺	0.11*
pre-net: District							
Left	-0.64*	-0.48*	0.45	-0.45***	-0.29	-0.45*	-0.04
Center-left	-0.09	-0.80	-0.48	0.73*	0.18	-0.74	0.30
Right	1.00***	1.36***	0.42	0.41***	0.08	0.23	0.88*
Reserved seats	-0.07	0.13	-0.03	-0.27 ⁺	0.30	-0.03	0.19
Others	1.98 ⁺	--	-1.92**	--	-0.29	0.26	--
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Same status, by bloc of the pair	C1	C2	C3	C4	C5	C6	C7
— pre-net: Lawyer							
Left	-0.07	0.06	0.02	0.06 ⁺	0.00	0.11*	-0.07
Center-left	0.11	0.11	0.13	-0.03	0.15*	-0.26	-0.00
Right	-0.11	-0.15**	-0.01	0.01	0.10	0.13*	-0.06
Reserved seats	0.06	0.24	0.10	-0.26 ⁺	0.20*	-0.21	-0.04
Others	0.12	-0.11	-0.09	0.09 ⁺	-0.00	-0.13	-0.15
Across blocs	0.06	-0.08*	0.02	0.04*	-0.05 ⁺	0.07 ⁺	0.06 ⁺
pre-net: Prior institutional experience							
Left	0.12 ⁺	0.13*	0.09	0.11**	0.04 ⁺	0.06	0.00
Center-left	0.11	-0.03	0.09	0.09 ⁺	-0.09	-0.01	0.12
Right	-0.13	-0.14	-0.03	0.05	-0.14	0.00	-0.11
Reserved seats	0.06	0.00	-0.08	-0.05	-0.11	-0.10	-0.05
Others	-0.25	-0.26	—	0.05	—	-0.20	-0.30
Across blocs	0.04	0.06 ⁺	0.03	0.06**	0.04*	0.04	0.04
— pre-net: Gender							
Left	-0.12	-0.03	-0.06	-0.03	-0.02	-0.04	-0.09
Center-left	-0.08	-0.10	-0.07	0.00	0.02	-0.02	0.05
Right	0.14*	0.03	0.05	-0.06	0.04	-0.03	0.09
Reserved seats	-0.11	-0.50	-1.77	-0.60**	-1.64***	0.04	-0.26
Others	-0.21	-0.26	-0.61	-0.20**	-0.38*	0.19	-0.37
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— pre-net: District							
Left	-0.64*	-0.48*	0.45	-0.45***	-0.29	-0.45*	-0.04
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Right	1.00***	1.36***	0.42	0.41***	0.08	0.23	0.88*
Reserved seats	-0.07	0.13	-0.03	-0.27 ⁺	0.30	-0.03	0.19
Others	1.98 ⁺	—	-1.92**	—	-0.29	0.26	—
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Left	−0.07	0.06	0.02	0.06 ⁺	0.00	0.11 [*]	−0.07
Center-left	0.11	0.11	0.13	−0.03	0.15 [*]	−0.26	−0.00
Right	−0.11	−0.15 ^{**}	−0.01	0.01	0.10	0.13 [*]	−0.06
Reserved seats	0.06	0.24	0.10	−0.26 ⁺	0.20 [*]	−0.21	−0.04
Others	0.12	−0.11	−0.09	0.09 ⁺	−0.00	−0.13	−0.15
Across blocs	0.06	−0.08 [*]	0.02	0.04 [*]	−0.05 ⁺	0.07 ⁺	0.06 ⁺
— pre-net: Prior institutional experience							
Left	0.12 ⁺	0.13 [*]	0.09	0.11 ^{**}	0.04 ⁺	0.06	0.00
Center-left	0.11	−0.03	0.09	0.09 ^{**}	−0.09	−0.01	0.12
Right	−0.13	−0.14	−0.03	0.05	−0.14	0.00	−0.11
Reserved seats	0.06	0.00	−0.08	−0.05	−0.11	−0.10	−0.05
Others	−0.25	−0.26	—	0.05	—	−0.20	−0.30
Across blocs	0.04	0.06 ⁺	0.03	0.06 ^{**}	0.04 [*]	0.04	0.04
— pre-net: Gender							
Left	−0.12	−0.03	−0.06	−0.03	−0.02	−0.04	−0.09
Center-left	−0.08	−0.10	−0.07	0.00	0.02	−0.02	0.05
Right	0.14 [*]	0.03	0.05	−0.06	0.04	−0.03	0.09
Reserved seats	−0.11	−0.50	−1.77	−0.60 ^{**}	−1.64 ^{***}	0.04	−0.26
Others	−0.21	−0.26	−0.61	−0.20 ^{**}	−0.38 [*]	0.19	−0.37
Across blocs	0.04	0.12 ^{**}	0.20 [*]	0.13 ^{***}	0.12 ^{***}	0.08 ⁺	0.11 [*]
— pre-net: District							
Left	−0.64 [*]	−0.48 [*]	0.45	−0.45 ^{***}	−0.29	−0.45 [*]	−0.04
Center-left	−0.09	−0.80	−0.48	0.73 [*]	0.18	−0.74	0.30
Right	1.00 ^{***}	1.36 ^{***}	0.42	0.41 ^{***}	0.08	0.23	0.88 [*]
Reserved seats	−0.07	0.13	−0.03	−0.27 ⁺	0.30	−0.03	0.19
Others	1.98 ⁺	—	−1.92 ^{**}	—	−0.29	0.26	—
Across blocs	0.54 ^{***}	0.28 [*]	0.67 ^{**}	0.39 ^{***}	0.47 ^{***}	0.34 [*]	0.46 ^{**}

	C1	C2	C3	C4	C5	C6	C7
— Controls							
Edges (baseline)	0.16	-0.49	-0.19	-0.28	-0.68	-0.55	-0.77
Commission member	0.79***	1.60***	0.87***	1.06***	1.66***	1.96***	2.20***
Range of θ_1 per doc (global)	-2.06***	-2.51***	-3.85**	-3.07***	-0.09	-1.89*	-1.55
Range of θ_2 per doc	-2.10***	-1.49*	-1.73	-2.35***	-2.02**	-1.49 ⁺	-0.71
— Bloc internal spread (θ_1 range of the own contingent, per doc)							
Left	-1.21	-1.16	-2.75	-3.70***	-3.81***	0.40	-1.80
Center-left	-1.76	0.23	0.23	0.16	-2.07	3.35 ⁺	-1.54
Right	0.20	-0.30	-5.29	-5.33*	-6.92	-0.45	-3.03
Reserved seats	-0.87	0.94	3.26	-1.40	-5.43 ⁺	0.15	0.93
Others	-4.82	-2.71 ⁺	0.55	-2.03**	-2.81***	-4.73*	-0.92
— Pre-network variables and structure							
Range of age per doc	-0.26	0.16	-0.20	0.03	-0.57**	0.01	0.17
Range of education per doc	-0.27	-0.57**	-0.28	0.04	-0.39*	-0.33	-0.02
Serial signers (gubdegree)	-0.63	-0.85	-0.05	-6.04*	-1.88***	1.22*	-0.55
Repeated pairs (gwdsp)	-0.42***	-0.28***	-0.38***	-0.16***	-0.23***	-0.35***	-0.24***

RQ2 — People

Does the network move positions (RQ2a) or behavior (RQ2b)?

RQ2a design

- Only the **norms era** (Feb 15 – May 14, content voted under the 2/3 rule).
- Exposure is the weighted mean position of your co-signers, where network weights decay over time ($m = 1 \approx 2$ wks, $m = 2 \approx 4$ wks, $m = 3 \approx 6$ wks).

$$E_{i,t}^{(m)} = \frac{\sum_{j \neq i} w_{ij,t}^{(m)} \theta_{j,t}}{\sum_{j \neq i} w_{ij,t}^{(m)}}, \quad w_{ij,t}^{(m)} = \sum_{s \leq t} \lambda_m^{t-s} \Delta w_{ij,s}$$

$$\Delta \theta_{i,t} = \alpha_i + \mu_t + \beta \theta_{i,t-1} + \lambda E_{i,t-1}^{(m)} + \lambda_f \tilde{E}_{i,t+2}^{(m)} + \varepsilon_{it}$$

- $\tilde{E}_{i,t+2}^{(m)}$: decay-weighted average of *future* exposures.

The results

Variable	$m = 1$	$m = 2$	$m = 3$
Lagged position $\theta_{i,t-1} (\hat{\beta})$	-0.756***	-0.756***	-0.756***
Decayed exposure $\hat{\lambda} (E_{i,t-1}^{(m)})$	0.022***	0.022***	0.021***
Decayed innovation $\hat{\lambda}_f$	-0.008	-0.007	-0.005
Wave-date fixed effects	yes	yes	yes
N (person-waves)	2,011	2,011	2,011
R^2 within	0.370	0.370	0.369

- The lagged exposure is significant across all memories m .
- The **future predicts nothing**, at every memory.
- The **influence component** survives every test; but latent homophily cannot be excluded by design.

RQ2b: discipline — the design

- When someone breaks ranks: alone, or with the people they wrote initiatives with?

$$X_{iv} = \frac{\sum_{j \neq i} w_{ij} D_{jv}}{\sum_{j \neq i} w_{ij}}, \quad \Pr(D_{iv} = 1) = \Lambda(\eta_i + \mu_{b(i),v} + \phi X_{iv} + \gamma \text{margin}_{iv})$$

- η_i absorbs the born rebels; $\mu_{b(i),v}$ absorbs “this vote split this bloc”.
- margin_{iv} : outsiders break ranks more.
- ϕ is the **parameter of interest**: do I defect more when the people I wrote initiatives with defect?

RQ2b: the results

	(1)	(2)
Exposure to defecting co-signers (ϕ)	8.82*** (0.60)	8.75*** (0.61)
Marginality in own bloc (γ)	—	1.09*** (0.09)
Person Fixed effects	yes	
Bloc $\times$ vote Fixed effects	yes	
N (member-votes)	186,988	

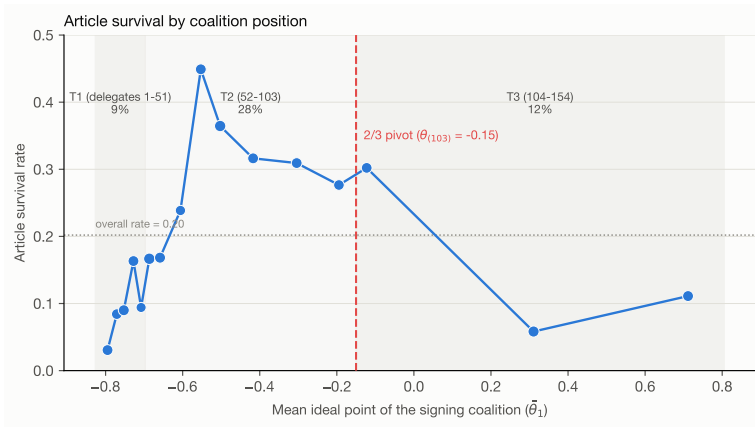
- The members whose co-signers defect, **defect far more**.
- Marginality predicts defection strongly but it's an **independent channel**.
- Defection travels along co-sponsorship ties.

RQ3a — Texts

What makes an article survive?

What survives?

- We have 1,565 *articles*, and 20.2% of them reach the draft — unevenly:



The survival model: article-level, complete

$$\Pr(\text{survives}_a) = \Lambda\left(\alpha_c + \gamma_1 |\bar{\theta}_{1,S_a} - \theta_{1,(103)}| + \gamma_2 sd(\theta_{1,S_a}) + \gamma_3 |S_a| + \gamma_4^\top C_{S_a} + \gamma_5^\top H_{S_a}\right)$$

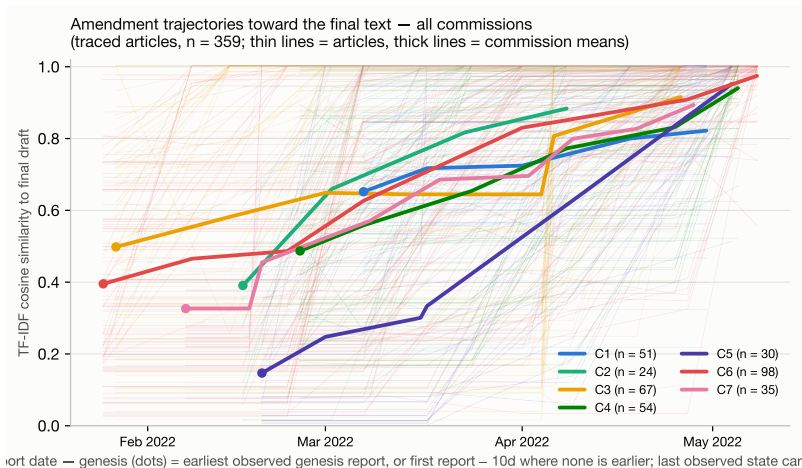
	(1) Pivotal	(2) + Network	(3) + Human capital
Coalition's distance to the $\frac{2}{3}$ pivot	-0.35**	-0.48**	-0.61**
Ideological heterogeneity $sd(\theta_1)$	0.27**	0.46**	0.53***
Coalition size	0.20	0.18	0.15
Mean betweenness		-0.22 ⁺	-0.23 ⁺
Mean constraint		0.01	0.26
Internal density (pairs with history)		0.25*	0.32**
Share of lawyers			-0.09
Share with prior experience			-0.16
Mean academic degree			-0.06
AIC	1393	1384	1386

RQ3b — Members

Does the context an initiative is born into matter?

Does the context an initiative is born into matter?

- **Success:** y_i = mean textual retention of i 's articles, 0 to 1.



Does the context an initiative is born into matter?

- Success is **not individual**: it clusters on the co-sponsorship network (Moran's $I = 0.44$, $p < .001$). Your success looks like your co-signers' success.
- To test whether the context matters, we ran the SDM (spatial Durbin model):

$$y = \rho W y + X\beta + W X\gamma + \varepsilon,$$

where W is the row-normalized co-sponsorship network and ρ answers whether that company matters.

	Own (β)	p	Co-signers' (γ , lag)	p
— Activity and network position				
Initiatives signed	-0.0001	.28	+0.0002	.78
Degree	+0.0004	.30	+0.0052	.31
Betweenness	-0.0000	.98	-0.0020	.40
— Ideological position				
Mean revealed vote (θ)	+0.008	.33	+0.056	.11
SD of θ	-0.018	.42	+0.026	.85
Distance to the $\frac{2}{3}$ pivot	-0.002	.95	-0.265	.050
Ego heterophily	-0.024	.088	-0.255	.055
— Credentials and profile				
Lawyer	+0.011	.13	+0.193	.041
Prior experience	+0.019	.031	+0.170	.11
Academic degree	-0.003	.49	-0.154	.012
Female	+0.002	.78	+0.208	.075
Age	-0.0002	.53	-0.0033	.40
— Coupling				
ρ^*	0.891	<.001	—	—

* Up to an identification test, ρ is the coalition-effectiveness spillover (Battaglini et al., AJPS 2020).

Conclusions

- The network **was predictable** from what people brought with them: political bloc and shared district (Right wing). Credentials **organized nothing**.
- $gwdsp < 0$ everywhere: coalitions re-form with fresh pairs. The structural signature of a **tabula rasa** (?).
- Political positions **drift toward the neighborhood**: an influence component, not just selection.
- Defection **travels along co-sponsorship ties**: when you break ranks, you do it with your co-signers.
- Articles win by geometry and team history, **not credentials**.
- Success is a **coalition good**. Your network shapes your success.

Thank you!



GitHub project repository

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